Ismail Aissaoui
Chlef, Algeria
Phone: +213 660 70 77 96
Email: ismail.aissaoui.pro@gmail.com

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Hiring Committee
Institute of Theoretical Computer Science
TU Graz
Graz, Austria

To the Hiring Committee,

I am writing to express my strong interest in the PhD Position in Machine Learning and Computational Neuroscience at TU Graz. As an AI & Data Science Engineer with a deep foundation in physics-informed neural networks and complex system modeling, I am highly motivated to contribute to your research on brain-inspired computing.

My engineering background has extensively focused on bridging biological/physical systems with advanced deep learning. In my recent "Cardiovascular Medical Digital Twin," I developed a 0D Windkessel **PI_DeepONet** that utilizes continuous regularization via PyTorch Autograd to enforce physical constraints. This rigorous mathematical approach to modeling fluid dynamics and biological states translates directly to the challenges of computational neuroscience, where modeling the intricate dynamics of brain networks requires high-fidelity, physics-informed architectures.

Furthermore, my implementation of **Medical_MambaTS**—a state space model designed for sub-millisecond processing of high-frequency temporal sequences (100Hz VitalDB telemetry)—demonstrates my ability to design scalable algorithms capable of handling the temporal complexities inherent in brain-inspired computing and neural signal processing.

I am eager to apply my strong programming skills and deep learning expertise to advance the theoretical and computational neuroscience initiatives at TU Graz. Enclosed are my CV and academic transcripts.

Sincerely,

Ismail Aissaoui